1. Overview

The Snatchforce Calculator estimates peak parachute opening loads (“snatch forces”) for a rocket’s drogue, main, and payload recovery systems. The program uses the standard drag-area snatch-force model: $F_{max} = Ck \cdot q \cdot Cd \cdot S_{proj}$ (Variables explained below)

The truth is, you can’t “calculate” a Ck value, especially without data, you can only choose one that best fits a given situation. Now, how does one choose a value?

- If you have flight-test or manufacturer data:
  Use the Ck value they provide or compute it from measured peak forces:
  - $Ck = \frac{F(measured)}{q \cdot Cd \cdot S_{proj}}$
  - $F(measured)$ is the peak (snatch) force measured when the parachute deploys in Newtons.
  - Ck is the opening-shock coefficient. It is a dimensionless multiplier describing how violently the parachute inflates.
    - $Ck = 1.0$ → force equals steady-state drag (very soft opening
    - $Ck > 1.0$ → realistic openings with inflation overshoot
    - Ck typically ranges 1.0–2.0 for student hemispherical chutes
    - Ck is not calculated from geometry – it is chosen from empirical data or standardized ranges.
  - q is dynamic pressure at line stretch
    - $q = \frac{1}{2}\rho V^2$
    - Where ρ is air density at deployment altitude
    - And V is line-stretch velocity (speed of the rocket right before the canopy begins to take load)
    - Units are **Pa** (N/m^2)
  - Cd is the steady-state drag coefficient of the parachute, based on projected area.
    - A hemispherical nylon canopy typically uses $Cd \approx 2.0$ to 2.3
    - Units: none (dimensionless).
  - S_{proj} is the projected area of the parachute canopy.
    - For a circular canopy: $S_{proj} = \pi\left(\frac{D}{2}\right)^2$
    - Where D is parachute diameter
    - Units: m^2
    - This is not the fabric surface area – it’s the 2-D circle footprint the air “sees.”

- If you do NOT have flight test data (most common):
  Use the ranges in the table above.
    - Use Lower Bound to represent the softest possible opening.
    - Use Estimated for design work.
    - Use Upper Bound for structural sizing and safety factor selection.

- If your chute is reefed or uses a slider:
  Reduce Ck (gentler inflation).

- If your chute deploys at high speed:
  Increase Ck (more violent inflation).

2. Input Units

- Choose a unit system at the top of the window:
    - **Metric** (kg, m, m/s)
    - **Imperial** (lb, ft, mph)
- All calculations internally convert to SI, and forces are output in **Newtons (N)**.

3. Input Fields (Per Chute)

- Each chute–**Drogue**, **Main**, and **Payload**–has its own input panel.
- Fill out the following fields for each chute as appropriate for your vehicle.

Mass

- Mass of the section attached to that parachute at deployment.
    - Drogue: full rocket mass
    - Main: descent mass after separation
    - Payload: payload-only mass
- Units: kg (metric) or lb (imperial)

Deploy Alt

- The altitude at which the chute is intended to deploy.
- Units: meters or feet

Deploy Speed

- Speed of the rocket when a chute deploys.
  - For Drogue, enter apogee separation speed or a small tumbling value.
  - For **Main** and **Payload**, you may check **"Calc from terminal velocity"**.
    When enabled, the program computes descent speed from the drogue chute using:
    $$V = \sqrt{\frac{2mg}{\rho C_D S_{proj,\,drogue}}}$$
- Units: m/s or mph (converted internally).

CoD (Drag Coefficient)

- Parachute drag coefficient based on **projected area**.
- Typical hemispherical nylon chute: **2.0–2.3**

Diameter

- Fully inflated canopy diameter.
- Units: m or ft
- Used to compute projected area:
  $$S_{proj} = \pi\left(\frac{D}{2}\right)^2$$

4. Choosing Ck (Opening-Shock Coefficient)

- You may supply **one exact Ck** or a **range** consisting of:
  - **Ck min** (lower bound: softest plausible opening)
  - **Ck est** (typical opening severity)
  - **Ck max** (upper bound: worst-case hard opening)
- **Recommended Ck Ranges for Hemispherical Student Rocket Chutes**
  - **Drogue**
    - Min: **1.0**
    - Est: **1.2**
    - Max: **1.4**
  - **Main**
    - Min: **1.0**
    - Est: **1.5**
    - Max: **2.0**

- **Payload**
  - Min: **1.0**
  - Est: **1.25**
  - Max: **1.5**

➢ These values come from experimental data and accepted high-power/IREC recovery guidelines.

➢ If "Exact Ck" is selected, only a single force value is computed.

5. Output Fields

➢ For each chute, the program computes and displays:

➢ **ρ (air density)**
  - Computed using the International Standard Atmosphere model.

➢ **q (dynamic pressure)**
  - $q = \frac{1}{2}\rho V^2$

➢ **Sproj (projected area)**
  - $S_{proj} = \pi\left(\frac{D}{2}\right)^2$

➢ **Fsteady**
  - Steady-state drag force:
  - $F_{steady} = qC_D S$

➢ **Snatchforce Results**

➢ Depending on your Ck selection:
  - **Force (min)** : using Ck min
  - **Force (est)** : using Ck est
  - **Force (max)** : using Ck max
  - or a single **Force** if Exact Ck is selected

➢ All forces are presented in **Newtons (N)**.

6. Compute Button

➢ Press **"Compute Snatchforces"** to calculate all drogue, main, and payload forces simultaneously. Each chute's results update immediately beneath its input panel.

7. Saving the Report

- ➢ Press **“Save Report to File”** to generate a text file containing:
  - all input values
  - computed air density
  - dynamic pressure\
  - projected area
  - chosen Ck values
  - minimum / estimated / maximum snatch forces
- ➢ The report is formatted for inclusion in IREC documentation and RSO review packages.

8. Interpretation

- ➢ Use:
  - **Force (max)** for **structural sizing** and safety-factor calculations
  - **Force (est)** for **nominal landing-load models**
  - **Force (min)** to understand gentlest-case conditions
- ➢ Hardware (e.g., U-bolts, harness, bulkheads) should be sized at **SF ≥ 2** relative to **Force (max)**.

9. How to Build the Program

If it is desired to modify the source program, one must know how to build the modified program. It is fairly straightforward to do.

1. The user must download and install QT Creator
2. Open QT Creator
3. Click the following:
   File → Open File or Project → Navigate to <Source Code Folder> → Double click “CMakeLists.txt”
4. Save the .cpp files that contain the source code
5. Run in QT Creator to test
6. Build when program finished